PROJECT REPORT

Title: Train Delay Predictor Using Machine Learning

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1.Introduction

This project focuses on predicting train delays using Machine Learning. Train delays affect passengers, scheduling, and logistics. By analysing factors like distance, weather, congestion, train type, time of day, and day of week, the model predicts whether a train will be delayed by more than 60 minutes. The system also provides an interactive command-line interface where users can enter journey details and get an instant prediction.

2.Problem Statement

The goal is to predict whether a train will be delayed by more than 60 minutes based on the following factors:

- Distance of the journey (in km)
- Weather conditions (Clear, Rainy, Foggy)
- Day of travel (Weekday or Weekend)
- Time of departure (Morning, Afternoon, Evening, Night)
- Type of train (Express, Superfast, Local)
- Route congestion level (Low, Medium, High)

3.Objectives

- Build a binary classification model to predict train delays
- Preprocess and encode real-world categorical data
- Train the model using Logistic Regression
- Scale features using StandardScaler

- Evaluate model accuracy on unseen test data
- Build an interactive CLI for live predictions

4.Literature / Research

Studies show that train delays depend on weather conditions, congestion levels, and operational factors such as train type and time of travel. Machine learning algorithms like Logistic Regression are commonly used as baseline classifiers for binary prediction tasks due to their simplicity and interpretability. Feature scaling is important for Logistic Regression as it is sensitive to the magnitude of input values.

5.Dataset Description

The dataset contains 2878 rows and 7 columns. Each row represents one train journey.

- distance — Journey distance in kilometres (range: 50 to 955 km)
- weather — Weather condition: Clear, Rainy, Foggy
- day — Day of journey: Monday to Sunday
- time — Departure time: Morning, Afternoon, Evening, Night
- train_type — Type of train: Express, Superfast, Local
- congestion — Route congestion: Low, Medium, High
- delay_min — Actual delay in minutes (range: 2 to 1230 minutes)

6.Methodology

Steps followed in this project:

1. Load the CSV dataset using pandas
2. Rename columns to lowercase for consistency
3. Encode all categorical columns into numbers using fixed mappings
4. Remove rows with missing values using dropna()
5. Create binary target variable from delay_min column
6. Select 6 input features and define target (y)
7. Split data: 60% training, 40% testing (random_state=50)
8. Scale features using StandardScaler

9. Train Logistic Regression model

10. Evaluate accuracy on test set

11. Accept user input and predict delay via CLI

7.Data Preprocessing

All categorical columns were converted to numbers using fixed mappings:

- Weather: Clear = 0, Rainy = 1, Foggy = 2
- Day: Monday to Friday = 0 (Weekday), Saturday & Sunday = 1 (Weekend)
- Time: Morning = 0, Afternoon = 1, Evening = 2, Night = 3
- Train Type: Express = 0, Superfast = 1, Local = 2
- Congestion: Low = 0, Medium = 1, High = 2

After encoding, rows with NaN values (from unmapped categories) were removed using dropna().

Target variable was created from delay_min:

```
dataset['delay'] = dataset['delay_min'].apply(lambda x: 1 if x > 60
else 0)
```

1 = Delayed (more than 60 minutes), 0 = On Time

8.Model Used

Algorithm: Logistic Regression (from scikit-learn)

- Train/Test Split: 60% training, 40% testing, random_state = 50
- Feature Scaling: StandardScaler was fit on training data only, then applied to test data
- Model was trained on 6 features: distance, weather, day, time, train_type, congestion
- delay_min column was NOT used as an input feature — only used to create the target

Code used:

```
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.4, random_state=50)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test  = scaler.transform(X_test)
model = LogisticRegression()
```

```
model.fit(X_train, y_train)
```

9.Results

The model was evaluated on the test set (40% of data):

- Accuracy: 0.7595486111111112 (approximately 75.95%)
- The model correctly predicts about 3 out of 4 journeys
- Remaining ~24% error is due to non-linear patterns that Logistic Regression cannot capture

10.Financial / Resource Analysis

- Software: Free — Python 3.x, pandas, scikit-learn (all open source)
- Hardware: Standard personal computer, no GPU needed
- Dataset: Free to download (CSV format)
- Cost: Minimal — no paid tools or services required

11.Challenges Faced

- Encoding categorical variables correctly — especially mapping days to weekday/weekend
- NaN errors appearing after .map() for any unrecognised category values
- Applying StandardScaler correctly — must fit only on training data to avoid data leakage
- Choosing the 60-minute threshold for the binary target variable
- Understanding why delay_min should not be used as an input feature

12.Future Scope

- Try advanced models: Random Forest, XGBoost for higher accuracy
- Add real-time weather data using APIs
- Include more features: station count, locomotive type, historical delay rates
- Build a web app using Flask or Streamlit
- Deploy as an API for integration with railway ticketing apps

13.Conclusion

This project successfully demonstrates how machine learning can be applied to predict train delays. A Logistic Regression model was trained on 2878 records using 6 features and achieved approximately 75.95% accuracy. The project covers the full pipeline from data loading and preprocessing to model training and interactive prediction via CLI. It serves as a strong baseline that can be extended with more advanced models and real-time data in the future.